VECTOR AI INSTALLATION GUIDE

Formal Implementation and Operational Guide

MAIC Nexus Challenge

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Table of Contents

CHAPTER 1 Introduction

1.1 Introduction

VectorAI platform is designed to support factories across different industries and is not limited to semiconductor manufacturing. The semiconductor factory used in the project is an example application used to demonstrate how the system may be applied to industrial equipment. The same overall approach can be applied to other manufacturing environments by changing the machine types, sensors, measurement parameters, operating limits and maintenance requirements according to the individual factory.

The overall connection can be understood as Machine → Sensor → Sensor Connection Kit → VectorAI Website. The machine is the physical equipment being monitored. The sensor measures a physical condition of the machine. The sensor connection kit receives and processes the sensor signal and transfers the measurement to VectorAI. The VectorAI website then presents the information in a form that can be understood and used by company personnel, maintenance teams, engineers and management.

1.2 Information Required Before Installation

Before any sensor is installed, the machine and monitoring requirements should be clearly identified. The factory should determine which machine is going to be monitored and why the machine requires monitoring. The objective may be to reduce unexpected downtime, identify abnormal machine behaviour, improve maintenance planning, monitor equipment health or provide management with better visibility of machine conditions.

Each machine should have a unique Machine ID. The Machine ID should remain consistent throughout the machine's use within the VectorAI platform. Information such as the machine name, machine type, production area, physical location, manufacturer and relevant operating information should also be recorded. This information is important because sensor measurements must always be associated with the correct machine.

The company should determine which physical condition of the machine needs to be measured. Depending on the machine, this may include temperature, vibration, pressure, force, motor load, current, flow, optical intensity or another measurable parameter. The selected measurement should have a meaningful relationship with the condition of the machine.

The sensor specification must be confirmed before installation. The selected sensor should have a suitable measurement range, accuracy, operating temperature, environmental rating, power requirement and installation method. Its output or communication interface must also be compatible with the sensor connection kit. Depending on the sensor, the connection may use an analog signal, digital signal, 4–20 mA, RTD, RS-485/Modbus, IO-Link, I^2C, SPI or another appropriate interface.

The physical installation location must be determined before the sensor is installed. The sensor should be placed at a location that provides a meaningful representation of the machine condition while remaining suitable for inspection and maintenance. Factors such as vibration, temperature, moisture, dust, accessibility, cable routing and possible electrical interference should be considered.

The factory should also determine the expected operating range of the measurement. A normal operating range, warning condition and critical condition should be defined where applicable. These limits should be based on manufacturer specifications, factory engineering knowledge, historical machine data or other validated information. Values used for the semiconductor example should not automatically be applied to other machines or industries.

CHAPTER 2 Step-by-Step Installation

2.1 Step 1: Select and Identify the Machine

The first step is to identify the physical machine that will be connected to VectorAI. The machine should be selected according to the factory's monitoring objectives and maintenance requirements. Machines that are important to production, have a history of downtime, or have measurable conditions that can provide early indications of failure may be suitable candidates for initial monitoring.

Once the machine has been selected, a unique Machine ID should be assigned. The machine name, machine type, production area and physical location should also be recorded. The company should identify the person or department responsible for the machine so that abnormal conditions detected by the system can be investigated appropriately.

The reason for monitoring the machine should be documented at this stage. For example, the factory may wish to monitor a production motor because overheating may result in unexpected downtime. Establishing this objective before selecting the sensor helps ensure that the sensor measures information that is genuinely useful to the factory.

2.2 Step 2: Determine What Needs to Be Measured

After identifying the machine, the next step is to determine which physical condition should be monitored. The selected parameter should provide useful information about the machine's operating condition.

For example, a motor may be monitored using temperature and vibration measurements. Temperature can provide information about overheating or abnormal loading, while vibration can provide information about mechanical conditions. A pressure-based machine may require pressure and temperature measurements, while another machine may require force, flow or optical measurements.

The company should determine how the measurement is expected to behave during normal operation. It should be established whether an increase, decrease, fluctuation or combination of conditions indicates an abnormal state. This information will later support the configuration of thresholds and the interpretation of sensor data.

2.3 Step 3: Select the Appropriate Sensor

Once the required measurement has been identified, an appropriate physical sensor should be selected. The sensor must be suitable for both the measurement and the environment in which it will operate. Important characteristics include measurement range, accuracy, response time, power requirements, environmental protection, mounting method and communication or output interface.

The selected sensor should be compatible with the sensor connection kit. If the sensor produces a signal that cannot be directly accepted by the connection kit, an appropriate interface or signal-conditioning component must be provided. The sensor manufacturer's datasheet and installation documentation should be retained for future reference.

The sensor models shown in the semiconductor example are reference examples rather than universal requirements. A different factory may use different sensor manufacturers or technologies while following the same VectorAI connection process.

2.4 Step 4: Determine and Prepare the Installation Location

The installation location should be determined carefully because measurement quality depends not only on the sensor itself but also on where and how it is installed. The sensor should be positioned so that it measures a condition representative of the machine's actual operating state.

The installation location should be reviewed for physical accessibility, available mounting space, cable routing, vibration, temperature, moisture, dust and other environmental conditions. The sensor should be securely mounted according to the manufacturer's instructions and should not be exposed to unnecessary mechanical stress or conditions outside its specified operating range.

The completed installation location should be documented. Where practical, a photograph, drawing or machine layout should be retained so that future maintenance personnel can identify the exact sensor position.

2.5 Step 5: Prepare the Sensor Connection Kit

Before connecting the sensor, the sensor connection kit should be prepared according to the selected sensor interface. The technician or engineering team should confirm the required supply voltage, connector configuration, wiring arrangement, signal type and input requirements.

Depending on the sensor, the connection kit may require an analog input, digital input, RTD interface, current-loop interface, RS-485/Modbus interface, IO-Link master or another appropriate module. All required interfaces should be verified before installation.

Power and communication cables should be properly organized and labelled. Cable routing should protect the wiring from mechanical damage, excessive heat, moisture and unnecessary electrical interference. The objective is to ensure that the sensor signal reaches the connection kit reliably and can be identified during future inspection or maintenance.

2.6 Step 6: Connect the Sensor to the Connection Kit

The sensor should then be physically connected to the appropriate input or interface of the sensor connection kit. The installation must follow the sensor manufacturer's instructions and the factory's applicable safety procedures.

Before power is applied, the wiring should be inspected carefully. The supply voltage, polarity, connector configuration and signal type should be checked against the sensor documentation. Where a communication-based sensor is used, its required communication settings should also be confirmed.

The connection should only be powered after the wiring has been verified. A record of the completed connection should be maintained, including the sensor identity, connection interface and relevant configuration information.

2.7 Step 7: Verify the Sensor Reading Locally

Before the sensor data is connected to the VectorAI website, the measurement should be verified locally at the sensor connection kit. This stage confirms that the physical sensor and its connection are functioning correctly before communication and website-related issues are introduced.

The local value should be checked for the correct unit, reasonable range, stability and response to changes in the physical condition. Where appropriate and safe, the measurement may be compared with a machine display or a trusted reference instrument.

If the reading is missing, physically impossible, unstable or significantly different from the expected condition, the problem should be resolved before proceeding. Possible causes include incorrect wiring, an unsuitable installation location, incorrect scaling, an incorrect interface configuration, sensor damage or calibration problems.

2.8 Step 8: Register the Machine and Sensor in VectorAI

To register a machine, the user first enters the Clean Room page (Figure 1) and places the required machine at its correct physical coordinate on the layout. The system then displays the machine information for the user to review and confirm (Figure 2).

After confirming the machine, the user selects SENSOR REGISTRY (NFC PAIRING) to open the sensor registration page (Figure 3). The user selects the appropriate sensor kit and uses the Tap and Scan function to scan the NFC information of the physical kit.

Once the kit is successfully detected, the user verifies the displayed information and selects Confirm and Bind. The system then binds the sensor kit to the selected machine, establishing the connection between the physical machine, sensor kit and VectorAI system (Figure 4).

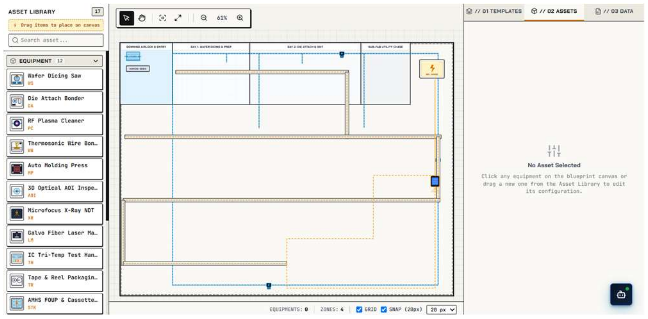

Figure 1: Initial clean room

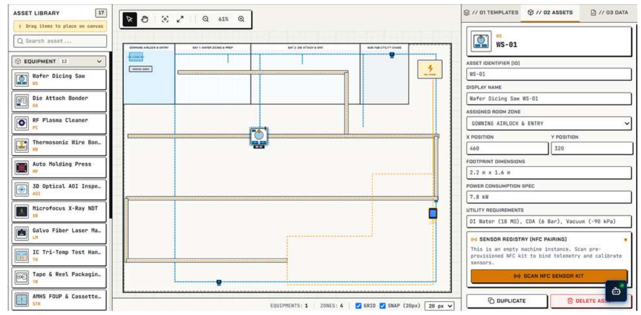

Figure 2: Adding machine in the clean room

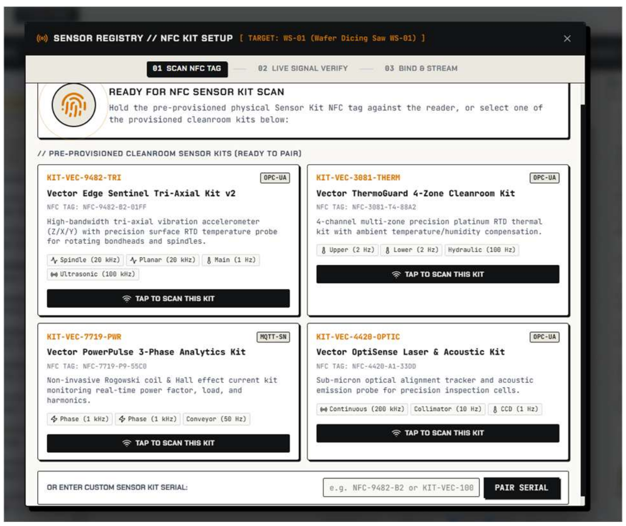

Figure 3: Scan NFC sensor kit page

Figure 4: Sensor kit detected and bind

2.9 Step 9: Transmit the Sensor Data to VectorAI

After the sensor kit has been successfully bound to the machine, the sensor can begin transmitting measurement data to VectorAI. The connection kit collects the sensor readings and sends the data to the platform using the configured communication method.

Each measurement should be associated with the correct Machine ID and Sensor ID and should include the measurement value, unit and timestamp. This allows VectorAI to correctly identify the source of each measurement and display the data under the appropriate machine. The purpose of this standardization is to separate the physical sensor technology from the website. The website should not need to understand the wiring or electronic characteristics of every sensor. Instead, it should receive a standardized measurement that can be interpreted consistently across different machines and factories.

2.10 Step 10: Verify the Measurement on the VectorAI Website

After data transmission begins, the user should return to the corresponding machine on the VectorAI website and verify that the sensor data is being received correctly. The displayed sensor value, unit, timestamp and sensor information should be checked against the expected reading from the sensor kit.

The measurement should appear under the correct machine and sensor. If the data is displayed correctly and updates as expected, the connection from the physical machine through the sensor kit to the VectorAI website can be considered successfully established.

2.11 Step 11: Configure Operating Thresholds

After the sensor connection has been successfully verified, the user can configure the operating thresholds for the selected sensor. The thresholds define the conditions under which the machine status is considered normal, requires attention or requires urgent action.

The normal range represents the expected operating condition, while warning and critical thresholds indicate increasing levels of abnormality. These values should be configured according to the specific machine, sensor characteristics and factory requirements. The thresholds should be reviewed and approved by the appropriate engineering, maintenance or operational personnel before being used for monitoring.

2.12 Step 12: Perform End-to-End Testing

The complete connection should be tested after the physical installation and VectorAI configuration have been completed. Testing should verify every stage of the system rather than confirming only that a number appears on the website.

The physical sensor should first be checked for correct operation. The connection kit should then be verified to ensure that it receives the expected value. The transmission process should be tested to confirm that the data reaches VectorAI with the correct Machine ID, Sensor ID, value, unit and timestamp. The website should then be checked to ensure that the measurement is displayed under the correct machine.

Normal, warning and critical conditions should be tested using safe and controlled test conditions wherever possible. Communication interruption should also be tested to ensure that missing data is not incorrectly displayed as normal operation. After communication is restored, the system should be checked to confirm that the sensor returns to the correct operating state.

CHAPTER 3 Technical Sensor and Connection Kit Installation

3.1 Sensor technical Specifications

Before installing a sensor, its technical specifications should be reviewed to ensure that it is suitable for the intended machine and measurement requirement. The sensor should be selected based on the physical parameter to be measured, such as temperature, vibration, pressure, current, force, flow or another relevant machine condition. Important specifications include the measurement range, accuracy, resolution, sampling rate, power supply requirement, operating temperature, environmental protection rating and communication or output interface.

The sensor's measurement range should be suitable for the expected operating condition of the machine. A sensor with an unsuitable range may produce inaccurate measurements or become damaged if the machine operates outside the sensor's allowable limits. The sensor should also be suitable for the factory environment, including factors such as temperature, humidity, dust, vibration and chemical exposure where applicable.

The sensor model, manufacturer, technical specifications and supporting documentation should be recorded as part of the installation documentation. This information provides a reference for future maintenance, replacement and troubleshooting activities.

CHAPTER 4 System Verification, Testing and Maintenance

4.1 Troubleshooting

If the website does not display a sensor value, troubleshooting should begin at the physical sensor and proceed through each stage of the connection. The sensor power supply, wiring, connector, interface configuration and local reading should first be examined. If the local reading is correct but the website does not receive the measurement, the communication and data-transmission path should then be investigated.

If the displayed value is incorrect, the unit, scaling, conversion, sensor configuration and Machine ID/Sensor ID mapping should be reviewed. If a measurement remains constant despite a known physical change, the sensor installation, sensor functionality and communication configuration should be examined.

Repeated warnings should not simply be removed by changing the threshold. The underlying machine condition, sensor condition, sensor location and configuration should first be investigated.

4.2 Testing After Maintenance or Sensor Replacement

Testing should also be performed whenever maintenance activities may affect the sensor, its mounting position, cabling or connection. Following maintenance, the local sensor reading should be verified and the VectorAI website should be inspected to confirm that the sensor remains correctly associated with the machine.

When a sensor is replaced, the new sensor should be registered with the correct Sensor ID and technical information. Its installation and calibration information should be recorded. A local reading test and a complete end-to-end test should then be performed. This prevents sensor replacement from creating incorrect or untraceable machine data.

4.3 Ongoing System Maintenance

Ongoing maintenance should cover both the physical sensor system and its digital configuration. During routine inspections, the sensor should be checked for physical damage, loose mounting, cable deterioration, connector problems, contamination or environmental effects.

Sensor readings should also be reviewed for unusual drift, repeated warnings, unexpected fluctuations or prolonged periods of unchanged values. Such behaviour may indicate either a real change in machine condition or a sensor and data-quality problem, and should therefore be investigated.

The VectorAI configuration should be reviewed periodically to ensure that machine identities, sensor identities, operating thresholds and other configuration information remain accurate. Calibration requirements should be followed according to the sensor manufacturer's recommendations and the factory's quality procedures.

4.4 Calibration and Data Quality

Calibration information should be maintained throughout the sensor's operational life. Calibration dates, calibration results and applicable next calibration dates should be recorded according to the factory's procedures and the sensor manufacturer's requirements.

Data quality should also be monitored continuously. The system should distinguish between a valid measurement and situations in which data is missing, invalid or unavailable. A communication failure should not automatically be interpreted as a normal machine condition.

If sensor drift or abnormal readings are detected, the issue should be investigated before modifying thresholds or machine-health rules. Maintaining reliable data is essential because VectorAI's analysis and maintenance information depends on the quality of the measurements received from the physical equipment.

4.5 Documentation and Records

A complete record should be maintained for every connected machine and sensor. The documentation should include the Machine ID, machine information, Sensor ID, sensor model, measurement purpose, installation location, wiring or interface information, connection-kit configuration and local test results.

The company should also retain the Machine-to-Sensor mapping, operating thresholds, calibration records, manufacturer documentation, end-to-end testing results, maintenance history and sensor replacement records. Maintaining these records provides traceability and makes future troubleshooting and maintenance significantly easier.

4.6 Final Acceptance

The connection should be considered ready for operational use only after the physical sensor, sensor connection kit, data transmission and VectorAI website have all been successfully verified.

The final acceptance process should confirm that the sensor is appropriate for the intended measurement, the installation has been properly documented, the local reading is reliable, the correct Machine ID and Sensor ID are used, the measurement reaches VectorAI correctly and the website displays the measurement under the correct machine.

The system should also demonstrate appropriate behaviour under normal, warning, critical and communication-loss conditions. All installation records, sensor specifications, calibration information, threshold definitions and test results should be retained as part of the factory's operational documentation.

CHAPTER 5 Conclusion

The complete VectorAI connection process can be summarized as a continuous chain from the physical machine to useful digital information. The machine produces a physical condition that can be measured, the sensor captures that condition, the sensor connection kit receives and transfers the measurement, and VectorAI presents the resulting information through the website.

The purpose of the system is therefore not simply to display sensor numbers. Reliable sensor data provides the foundation for monitoring machine conditions, identifying abnormal behaviour, supporting maintenance activities and improving the factory's understanding of equipment health. Because the architecture is based on a general machine-to-sensor-to-platform approach, the same concept can be adapted to different factories and industries while using semiconductor manufacturing as only one example application.

CHAPTER 6 Appendix

Appendix A – Example Machine-to-Sensor Mapping

The following example illustrates how a physical machine and sensor can be represented in VectorAI. This does not define a universal factory configuration.

Item	Example	Purpose
Machine ID	MTR-001	Uniquely identifies the machine.
Machine Name	Production Motor 01	Provides a human-readable machine name.
Sensor ID	motor_temperature_01	Uniquely identifies the sensor.
Measurement	Motor Temperature	Defines what the sensor measures.
Unit	°C	Defines the measurement unit.
Current Value	42.3	Example live measurement.
Status	Normal	Example interpretation against approved operating limits.

Appendix B – Recommended Acceptance Test Record

Test Area	Expected Result	Actual Result	Status
Physical sensor installation	Sensor securely installed and suitable for measurement.		
Local sensor reading	Correct value and unit received by connection kit.		
Sensor identity	Correct Sensor ID and Machine ID.		
Data transmission	Measurement reaches VectorAI.		
Website display	Correct value appears under correct machine.		
Threshold behaviour	Normal, Warning and Critical states behave as configured.		
Communication loss	System indicates unavailable/offline data appropriately.		
Recovery	Data returns correctly after communication is restored.		